KU LEUVEN

FACULTY OF ENGINEERING SCIENCE

MASTER OF ARTIFICIAL INTELLIGENCE

Robotics

Homework B

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1 Q1

The homework defines the prior

$$p(x) = 0.5\,\mathcal{N}(x \mid -1.0,1) + 0.5\,\mathcal{N}(x \mid 1.0,0).$$

I keep the statement as written. I interpret the second term as a degenerate Gaussian, that is, a point mass at $x = 1$.

I sample 10000 prior values with random seed 42. These prior samples have mean -0.00800 and standard deviation 1.22944.

No extra normalization is needed. The mixture weights already sum to one.

The measurement model is

$$z = 2x + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,0.3), \qquad z = 0.5.$$

I update the prior with Bayes' rule:

$$p(x \mid z) \propto p(z \mid x)\,p(x).$$

As Fig. 1 shows, the prior has a broad Gaussian component around $x = -1$ and a spike at $x = 1$. The posterior shifts toward values that explain $z = 0.5$, so the continuous part moves toward $x \approx 0.25$. The point mass at $x = 1$ remains, but its weight decreases.

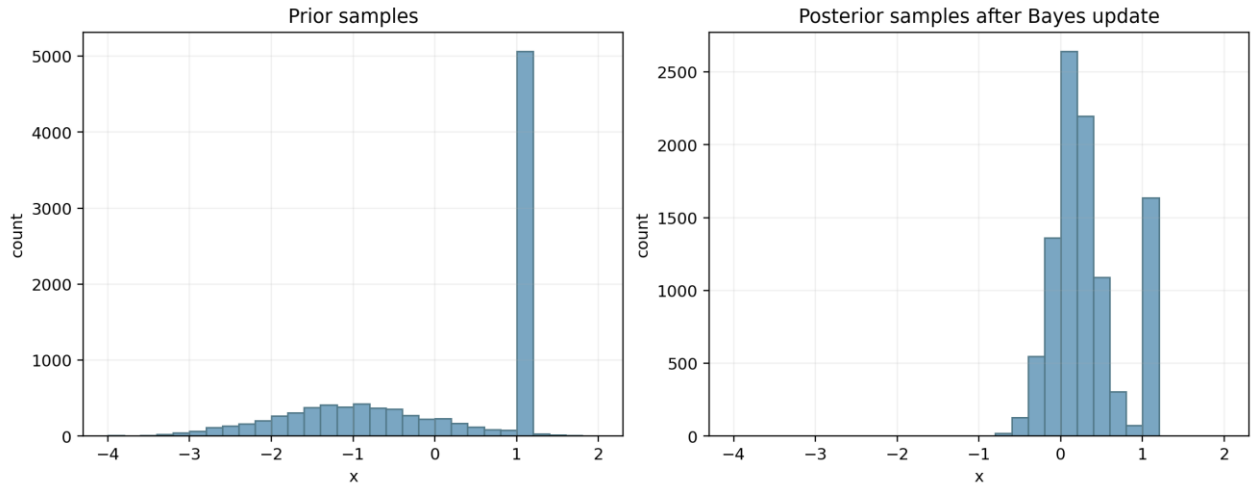

Fig. 1. Prior and posterior sample histograms for Homework B1.

The posterior sample mean is 0.30263, and the posterior standard deviation is 0.38992.

The resulting distribution is not a single Gaussian. It is a mixed distribution. More precisely, it is a Gaussian mixture with one degenerate component.

2 Q2

Let $J_p(q) \in \mathbb{R}^{3\times 6}$ denote the translational Jacobian of the end effector. Let

$$B = [u \;\; v] \in \mathbb{R}^{3\times 2}.$$

Because u and v span the plane, the orthogonal projector onto that plane is

$$P = B(B^T B)^{-1} B^T.$$

The in-plane velocity is

$$\dot{p}_\Pi = P J_p \dot{q}.$$

Under the standard manipulability condition $\|\dot{q}\|_2 \leq 1$, the set of all reachable in-plane velocities is

$$\mathcal{V}_\Pi = \{P J_p \dot{q} : \|\dot{q}\|_2 \leq 1\}.$$

This is the linear image of the unit ball in joint space. Hence the reachable set is an ellipse in the plane. It degenerates to a segment or a point if the in-plane rank drops.

The maximum velocity magnitude inside the plane is

$$v_{\max} = \sigma_{\max}(P J_p) = \sqrt{\lambda_{\max}(J_p^T P J_p)}.$$

This quantity is the major semi-axis length of the planar manipulability ellipse.

3 Q3

I use the lecture parameterization

$$\mu = [m \quad mc_x \quad mc_y \quad mc_z \quad I_{xx} \quad I_{xy} \quad I_{xz} \quad I_{yy} \quad I_{yz} \quad I_{zz}]^T.$$

3.1 (a) Transforming μ^a to μ^b

Let ${}^aT_b = ({}^aR_b, {}^ap_b)$ denote the rigid transform from frame b to frame a. I reconstruct m, c^a, and I^a from μ^a. Then I compute the central inertia

$$I_c^a = I^a - m[c^a \times]^T [c^a \times].$$

Next, I transform the center of mass and the central inertia to frame {b}:

$$c^b = ({}^aR_b)^T(c^a - {}^ap_b), \qquad I_c^b = ({}^aR_b)^T I_c^a \, {}^aR_b.$$

Finally, I apply Steiner's rule again:

$$I^b = I_c^b + m[c^b \times]^T[c^b \times].$$

Then I assemble μ^b from m, mc^b, and the entries of I^b.

3.2 (b) Physical realizability

The lecture writes the spatial inertia matrix as

$$\mathcal{M} = \begin{bmatrix} I & m[c \times] \\ m[c \times]^T & mI_{3\times 3} \end{bmatrix}.$$

The kinetic energy is

$$T = \frac{1}{2}[\omega^T \quad v^T]\mathcal{M}\begin{bmatrix} \omega \\ v \end{bmatrix}.$$

So μ is physically realizable if $m > 0$ and the associated spatial inertia matrix is symmetric positive definite:

$$\mathcal{M} \succ 0.$$

3.3 (c) Rigidly linked bodies

If two rigidly linked bodies are expressed in the same frame, their composite parameters add directly:

$$\mu^{ab} = \mu^a + \mu^b.$$

If the two bodies are given in different frames, I first transform one parameter vector into the frame of the other body, then I add the two vectors.

4 Q4

For an ideal contact, the twist t and wrench w satisfy the reciprocity relation

$$w^T t = \tau^T \omega + F^T v = 0.$$

This means that each task-frame direction is either force-controlled or motion-controlled. The environment determines the natural constraints. The controller determines the artificial constraints.

4.1 (a) Peg insertion

I choose the task frame with z_t along the peg axis and x_t, y_t in the lateral directions.

Natural constraints:

$$v_{x_t} = 0, \quad v_{y_t} = 0, \quad \omega_{x_t} = 0, \quad \omega_{y_t} = 0, \quad F_{z_t} = 0, \quad \tau_{z_t} = 0.$$

Artificial constraints:

$$F_{x_t},\ F_{y_t},\ \tau_{x_t},\ \tau_{y_t},\ v_{z_t},\ \omega_{z_t}.$$

One consistent task-frame specification is:

x_t:force 0
y_t:force 0
z_t:velocity v_{ins}
ax_t:torque 0
ay_t:torque 0
az_t:velocity 0

4.2 (b) Flat object on a plane

I choose z_t normal to the plane and x_t, y_t in the plane.

Natural constraints:

$$v_{z_t} = 0, \quad \omega_{x_t} = 0, \quad \omega_{y_t} = 0, \quad F_{x_t} = 0, \quad F_{y_t} = 0, \quad \tau_{z_t} = 0.$$

Artificial constraints:

$$v_{x_t},\ v_{y_t},\ \omega_{z_t},\ F_{z_t},\ \tau_{x_t},\ \tau_{y_t}.$$

One consistent specification is:

x_t:velocity v_x^d
y_t:velocity v_y^d
z_t:force F_n^d
ax_t:torque 0
ay_t:torque 0
az_t:velocity ω_z^d

4.3 (c) Wedge on a plane

I choose x_t along the contact line, y_t in the support plane and perpendicular to the line, and z_t normal to the plane.

Natural constraints:

$$v_{z_t} = 0, \quad \omega_{y_t} = 0, \quad F_{x_t} = 0, \quad F_{y_t} = 0, \quad \tau_{x_t} = 0, \quad \tau_{z_t} = 0.$$

Artificial constraints:

$$v_{x_t},\ v_{y_t},\ \omega_{x_t},\ \omega_{z_t},\ F_{z_t},\ \tau_{y_t}.$$

One consistent specification is:

x_t:velocity v_x^d
y_t:velocity v_y^d
z_t:force F_n^d
ax_t:velocity ω_x^d
ay_t:torque 0
az_t:velocity ω_z^d

4.4 (d) Twist-wrench relationship in any object-attached frame

Let {o} be any frame attached to the object and {t} the task frame. Then the screws transform as

$$t_o = \mathrm{Ad}_{{}^oT_t}\, t_t, \qquad w_o = \mathrm{Ad}_{{}^oT_t}^{-T} w_t.$$

The reciprocity relation is invariant:

$$w_o^T t_o = w_t^T t_t = 0.$$

So the admissible twist space and the admissible contact wrench space remain reciprocal in any object-attached frame.